MID TERM EXAMINATION

Paper: Mid Term Examination - Set A | Duration: 3 Hours | Max Marks: 100 | Date: 14-03-2026

Total Questions: 16 | *Answer all questions.*

Name: ____________________________ **Roll No:** _________________________

Student Sign: ___________________ **Teacher Sign:** ___________________

SECTION A: MCQS

1. Which of the following is NOT listed as a common pitfall in the design process due to a flawed process? **[2 Marks]**

a) No early analysis and understanding of the user's needs.
b) A focus on using design features that are "neat".
c) Extensive testing performed before product release.
d) No common design team vision of user interface design goals.

2. According to Nielsen (2003), which of the following is one of the five quality components of usability? **[2 Marks]**

a) Flexibility
b) Memorability
c) Attractiveness
d) Adaptability

3. In the context of human characteristics in design, which principle states that our eyes and mind see objects as belonging together if they are near each other in space? **[2 Marks]**

a) Similarity
b) Closure
c) Proximity
d) Continuity

4. What does Fitts' Law (1954) primarily state regarding target acquisition time in screen design? **[2 Marks]**

a) Time is a function of the user's age and experience.
b) Time is a function of the distance to and size of the target.
c) Time is inversely proportional to the user's manual dexterity.
d) Time is independent of the target's visual complexity.

5. Which of the following is a common psychological response to poor design, described as an overabundance of annoyances or an inability to easily convey intentions to the computer? **[2 Marks]**

a) Confusion
b) Boredom
c) Frustration
d) Panic or stress

SECTION A: SHORT ANSWERS

6. List three general observations about design made by Gould (1988) regarding the development process.
- Nobody ever gets it right the first time.
- Development is chock-full of surprises.
- Good design requires living in a sea of changes.
- Making contracts to ignore change will never eliminate the need for change.
- Even if you have made the best system humanly possible, people will still make mistakes when using it.
- Designers need good tools. **[5 Marks]**

7. Define the term "usability" as provided in the text, including the meaning of "easily" and "effectively."
Usability is defined as "the capability to be used by humans easily and effectively, where,
- easily = to a specified level of subjective assessment,
- effectively = to a specified level of human performance." **[5 Marks]**

8. Enumerate five common usability problems in graphical systems as reported by IBM usability specialists, according to Mandel (1994).
- Ambiguous menus and icons.
- Languages that permit only single-direction movement through a system.
- Input and direct manipulation limits.
- Highlighting and selection limitations.
- Unclear step sequences.
- More steps to manage the interface than to perform tasks.
- Complex linkage between and within applications.
- Inadequate feedback and confirmation.
- Lack of system anticipation and intelligence.
- Inadequate error messages, help, tutorials, and documentation. **[5 Marks]**

9. Briefly explain the difference between short-term (working) memory and long-term memory as described in the text.
- Short-term, or working, memory receives information from either the senses or long-term memory, usually cannot receive both at once, and information stored within it is thought to last from 10 to 30 seconds, with limited processing capacity.
- Long-term memory contains the knowledge we possess, stores information transferred from short-term memory through a complex learning process, has a capacity thought to be unlimited, and learning is improved if information has structure, is meaningful, familiar, and through repetition. **[5 Marks]**

10. List three physical characteristics of people that can affect their performance with a system.
- Age
- Hearing
- Vision
- Cognitive Processing
- Manual Dexterity
- Gender
- Handedness
- Disabilities **[5 Marks]**

SECTION A: LONG ANSWERS

11. Describe the design commandments that, if foremost in the designer's mind, can help eliminate pitfalls in designing for people.

The design commandments to eliminate pitfalls are:

- Provide a multidisciplinary design team including specialists in Development, Interface Design, Visual design, Usability assessment, Documentation, and Training.
- Solicit early and ongoing user involvement to gain direct knowledge about jobs, tasks, and needs, and to confront user resistance to change.
- Gain a complete understanding of users and their tasks, recognizing that the product must be geared to people's needs, not developers'.
- Create the appropriate design, considering the total user experience and as many alternative designs as possible before selection.
- Begin utilizing design standards and guidelines at the start of the design process.
- Perform rapid prototyping and testing continually during all stages of development to quickly identify and solve problems and uncover defects.
- Modify and iterate the design as much as necessary, establishing user performance and acceptance criteria and continuing testing until all design goals are met.
- Integrate the design of all system components including software, documentation, help function, and training, developing them concurrently. **[10 Marks]**

12. Discuss the five quality components of usability as suggested by Nielsen (2003).

Nielsen (2003) suggests usability possesses these five quality components:

- Learnability: How easy it is for users to accomplish basic tasks the first time they encounter the design.
- Efficiency: Once users have learned the design, how quickly they can perform tasks.
- Memorability: When users return to the design after a period of not using it, how easily they can reestablish proficiency.
- Errors: How many errors users make, how severe these errors are, and how easily they can recover from them.
- Satisfaction: How pleasant it is to use the design. **[10 Marks]**

13. Elaborate on how human perception, including various perceptual characteristics, influences design.

Perception is our awareness and understanding of environmental elements and objects through physical sensations. It is influenced by experience and several perceptual characteristics:

- Proximity: Objects near each other in space are seen as belonging together.
- Similarity: Objects sharing a common visual property (color, size, shape) are seen as belonging together.
- Matching patterns: Similar responses to the same shape in different sizes (e.g., alphabet letters).
- Succinctness: Objects are seen as having perfect or simple shapes because they are easier to remember.
- Closure: Perception is synthetic; if something doesn't quite close, we see it as closed anyway to establish meaningful wholes.
- Unity: Objects forming closed shapes are perceived as a group.
- Continuity: Shortened lines may be automatically extended.
- Balance: We desire stabilization or equilibrium; vertical, horizontal, and right angles are visually satisfying and easy to look at.
- Expectancies: Perception is influenced by what we expect to be there, leading to errors like missing spelling mistakes.
- Context: Surroundings and environment influence perception, making objects appear differently depending on adjacent elements or external comments.

- Signals versus noise: Our senses are bombarded by stimuli, distinguishing important signals from unwanted noise. **[10 Marks]**

14. Compare and contrast the characteristics of novice users versus expert users in business systems, according to the text.

Novice Users:

- Depend on system features that assist recognition memory: menus, prompting information, instructional and help screens.
- Need restricted vocabularies, simple tasks, small numbers of possibilities, and very informative feedback.
- View practice as an aid to moving up to expert status.
- Often have difficulties with dragging and double-clicking, distinguishing double-clicks from two separate clicks.
- Struggle with window management, not always realizing overlapping windows represent a three-dimensional space or that hidden windows still exist.
- Have difficulty with file management, especially with structures nested more than two levels deep, as structure is not as apparent as with physical files.
- Possess a fragmented conceptual model of a system.
- Organize information less meaningfully, orienting it toward surface features of the system.
- Structure information into fewer categories.
- Have difficulty in generating inferences and relating new knowledge to their objectives and goals.
- Pay more attention to low-level details and surface features of the system.

Expert Users:

- Rely upon free recall.
- Expect rapid performance.
- Need less informative feedback.
- Seek efficiency by bypassing novice memory aids, reducing keystrokes, chunking, and summarizing.
- Possess an integrated conceptual model of a system.
- Possess knowledge that is ordered more abstractly and more procedurally.
- Organize information more meaningfully, orienting it toward their task.
- Structure information into more categories.
- Are better at making inferences and relating new knowledge to their objectives and goals.
- Pay less attention to low-level details and surface features of a system. **[10 Marks]**

15. Explain how individual differences complicate design and what considerations should be made to address these differences.

Individual differences complicate design because there is no average user; people differ widely in looks, feelings, motor abilities, intellectual abilities, learning abilities and speed, among others. The design must permit people with widely varying characteristics to satisfactorily and comfortably learn the task or job, or use the Web site.

To address these differences:

- Design must provide for the needs of all potential users.
- Multiple versions of a system can be created.
- Consideration of computer literacy, system experience (novice vs. expert), application experience, task experience, other system use, education, reading level, typing skill, native language and culture.
- Consideration of mandatory vs. discretionary use, frequency of use, task or need importance, task structure, social interactions, job category, and lifestyle.
- Consideration of psychological characteristics such as attitude and motivation, patience, stress level, expectations, and cognitive style (verbal vs. spatial, analytic vs. intuitive, concrete vs. abstract).

- Consideration of physical characteristics such as age (young adults vs. older adults regarding computer use, reading speed, memory, and task completion), hearing, vision (font size, type, color contrast, link design), cognitive processing (slowing with age), manual dexterity, gender (strength, hand size, color-blindness), handedness, and disabilities (blindness, defective vision, color-blindness, poor hearing, deafness, motor handicaps). **[10 Marks]**

SECTION A: SHORT ANSWERS

16. Discuss the psychological and physical responses people exhibit to poor design in computer systems.
Poor design in computer systems can elicit various psychological and physical responses, significantly diminishing user effectiveness and efficiency.
PSYCHOLOGICAL RESPONSES:
- Confusion: Occurs when detail overwhelms perceived structure, making meaningful patterns difficult to ascertain and preventing the understanding or establishment of a conceptual model.
- Annoyance: Caused by roadblocks that prevent prompt and efficient task completion or need satisfaction. Examples include design inconsistencies, slow computer reaction times, difficulty finding information, outdated information, and visual screen distractions.
- Frustration: Results from an overabundance of annoyances, an inability to easily convey intentions to the computer, or inability to finish a task. It's heightened by unexpected, irreversible computer responses or inflexible/unforgiving systems.
- Panic or stress: Introduced by unexpectedly long delays during severe or unusual pressure, such as unavailable systems or long response times under deadlines or when dealing with irate customers.
- Boredom: Arises from improper computer pacing (slow response/download times) or overly simplistic jobs.

These psychological responses diminish user effectiveness by blocking concentration, forcing task-irrelevant thoughts, and leading to higher error rates, poor performance, anxiety, and dissatisfaction.
PHYSICAL RESPONSES: These often accompany or are direct consequences of psychological responses:
- Abandonment of the system: Users reject the system and rely on other information sources. Common for managerial/professional personnel and Web users who have this discretion.
- Partial use of the system: Only a portion of the system's capabilities are used, typically those easiest to perform or most beneficial. Historically, this has been the most common reaction, leaving many system aspects unused.
- Indirect use of the system: An intermediary is placed between the user and the computer. Typical for managers or those with authority due to its requirement of high status and discretion.
- Modification of the task: The user changes the task to match the system's capabilities. Prevalent when tools are rigid and the problem is unstructured, as in scientific problem-solving.
- Compensatory activity: Additional actions are performed to compensate for system inadequacies, such as manual reformatting of information. Common among workers with limited discretion like clerical personnel.
- Misuse of the system: Rules are bent to shortcut operational difficulties. Requires significant system knowledge and may affect system integrity.
- Direct programming: Sophisticated workers reprogram the system to meet specific needs.

These physical responses also significantly diminish user efficiency and effectiveness by forcing reliance on other sources, underutilization of capabilities, or time-consuming "work-around" actions. **[15 Marks]**

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